lncRNAs as cell type-specific biomarkers across neural and non-neural human tissues.

Sasha Becker[1,2], Bingfang Xu[1], Ajith V. Pankajam[1], Richard Scheuermann[1]

[1] Division of Intramural Research, National Library of Medicine, National Institutes of Health
[2] Department of Biostatistics and Bioinformatics, Milken Institute School of Public Health, George Washington University

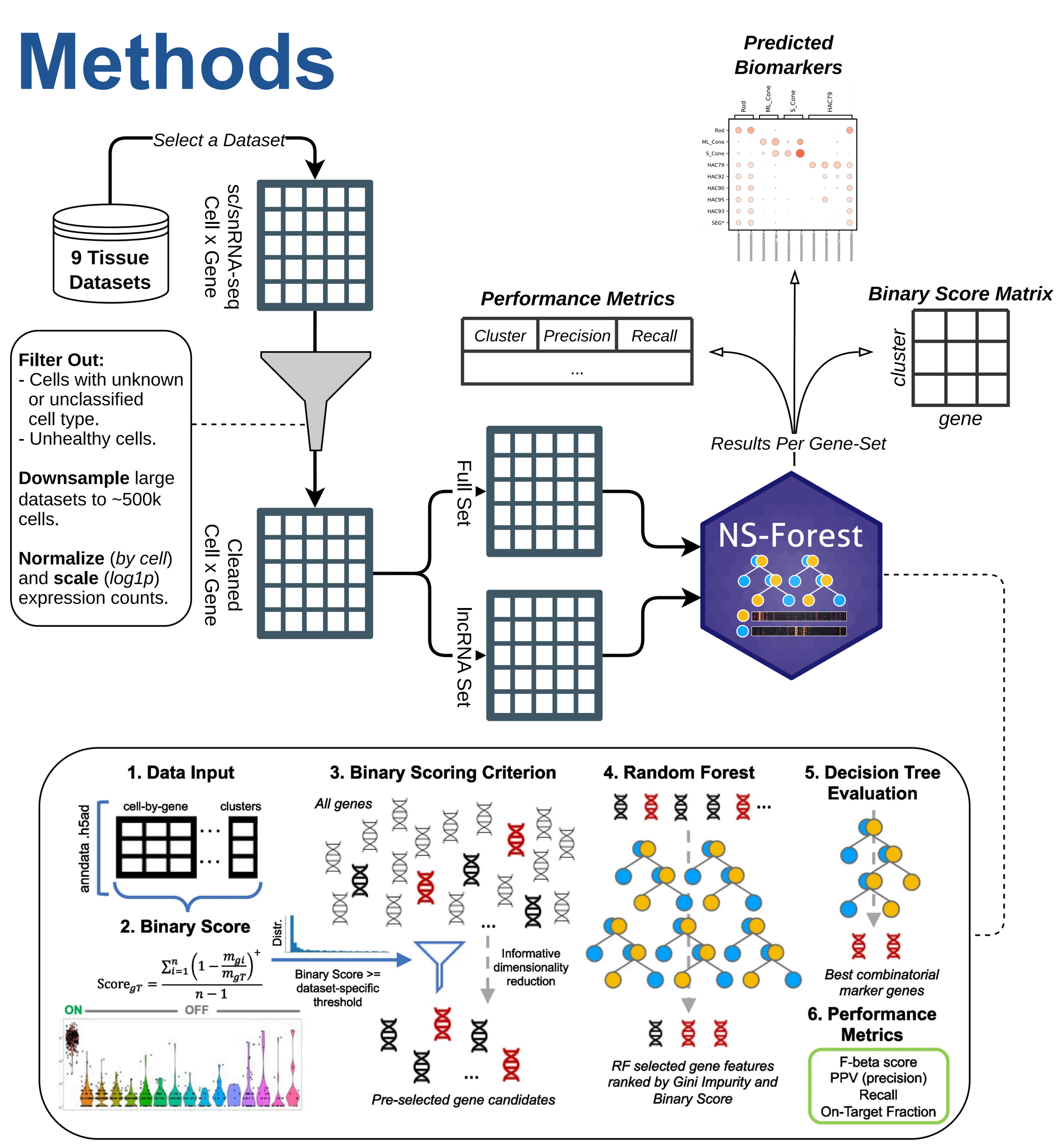
Introduction

Long non-coding RNA (lncRNA) are non-coding RNA transcripts longer than 200 nucleotides that regulate gene expression through diverse nuclear and cytoplasmic mechanisms.[1] Compared to protein-coding genes, lncRNAs generally exhibit **lower expression** but **greater cell-type specificity**,[2] making them promising candidate biomarkers for cellular identity. However, their utility for **distinguishing healthy cell types** across tissues remains **poorly characterized**.

NS-Forest v4.1[3] identifies minimal sets of **marker genes** that classify individual cell types using random forest models. Candidate biomarkers are prioritized based on **positive binary expression patterns**, quantified by NS-Forest's **Binary Score**, which measures the specificity of a gene's expression within a cell type. In running NS-Forest on diverse tissues, it **frequently identified lncRNAs** as cell type specific markers if the tissue was enriched in **neuronal cell types**.

We aim to evaluate whether lncRNAs alone retain sufficient discriminatory information to classify cell types and whether their biomarker potential differs between neural and non-neural healthy human tissues.

Methods



Results

Table 1. **Descriptive table of Cell x Gene data.**

Type	Tissue	Number of Cell Types	Number of Cells (filtered and downsampled)	Gene Set	Number of Genes	Number of Positive Genes (% of Set)	Mean Number of Genes Passing Threshold/Cell Type
Non-Neural	Bone Marrow[4]	45	66.6k	Full	49,082	6,474 (13.19%)	622.11
				lncRNA	12,128	79 (0.65%)	7.26
	Breast[5]	42	507.0k	Full	33,604	4,622 (13.75%)	425.64
				lncRNA	14,399	64 (0.44%)	5.11
	Kidney[6]	75	107.7k	Full	33,140	5,249 (15.84%)	465.61
				lncRNA	11,509	203 (1.76%)	12.81
	Liver[4]	29	259.6k	Full	35,659	3,073 (8.62%)	295.41
				lncRNA	11,560	41 (0.35%)	3.17
	Lung[7]	61	491.2k	Full	27,402	5,288 (19.30%)	461.93
				lncRNA	8,744	83 (0.95%)	4.78
Neural	Motor Cortex[8]	127	76.5k	Full	30,751	8,723 (28.37%)	651.53
				lncRNA	2,032	262 (12.89%)	22.03
	Middle Temporal Gyrus[9]	75	15.6k	Full	30,751	11,529 (37.49%)	750.94
				lncRNA	2,032	386 (19.00%)	32.65
	Retina[10]	123	509.0k	Full	35,475	6,925 (19.52%)	590.61
				lncRNA	15,491	703 (4.54%)	56.54
	Spinal Cord[11]	43	28.6k	Full	35,467	8,425 (23.75%)	515.18
				lncRNA	15,488	703 (4.54%)	61.62

- **Neural tissues** contain a **larger proportion of positive lncRNAs**, increasing the **contribution of lncRNAs** to candidate neural biomarkers pools and indicating **comparably higher lncRNA expression** in neural cell types.
- **Neural tissues** retain a **larger** average of **candidate lncRNA biomarkers** per cell type, reflecting **greater diversity** in cell type-specific **lncRNA expression**.

Figure 1. **Distribution of non-zero Binary Scores of a gene within a cell type.**

- There is **some separation in distribution for the neural cells** (δ = 0.228; lncRNA has higher Binary Scores) and **negligible separation** for the **non-neural** cells (δ = -0.032; full transcriptome has higher Binary Scores).
- In **neural tissues**, lncRNAs show a **greater proportion of high Binary Scores** compared with the full transcriptome, consistent with **greater cell type-specific expression**.
- In **non-neural tissues**, Binary Score distributions are **broadly comparable** between gene sets, suggesting that the two are **similarly specific**.

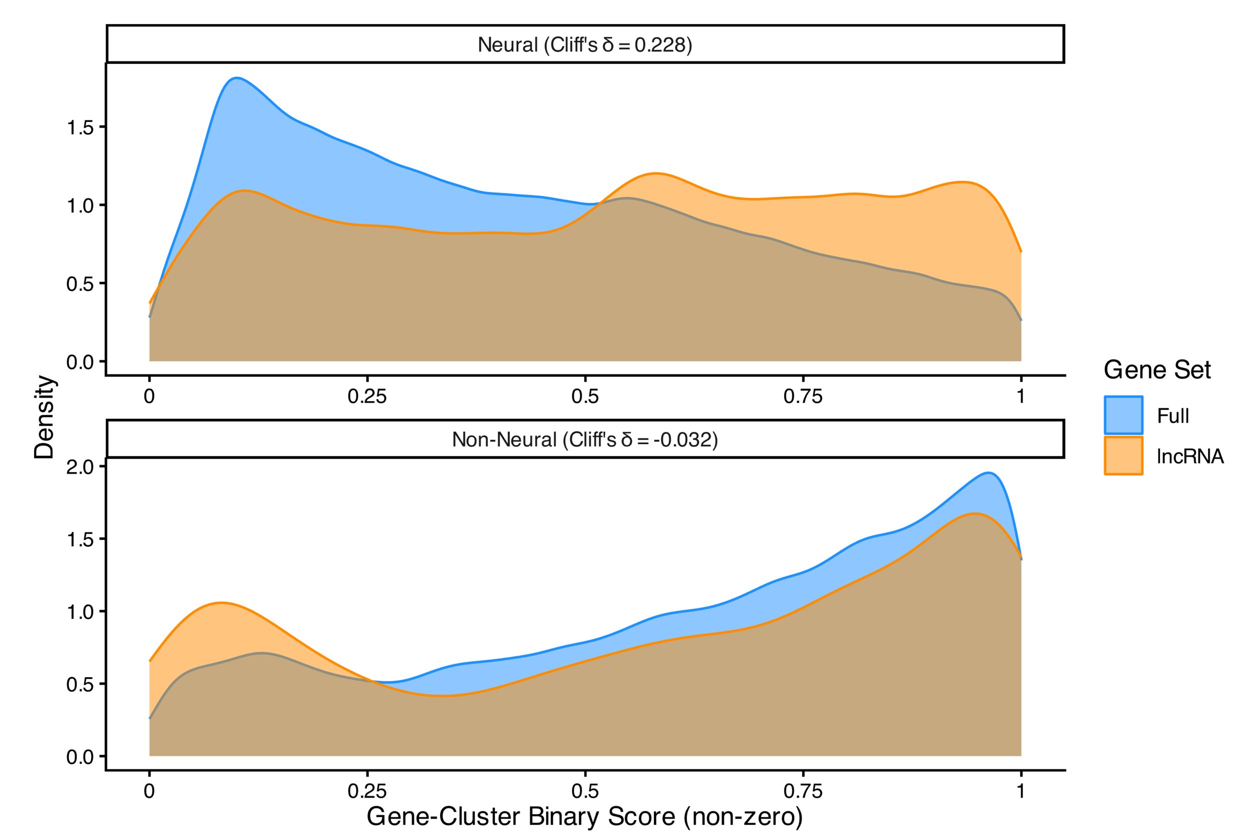
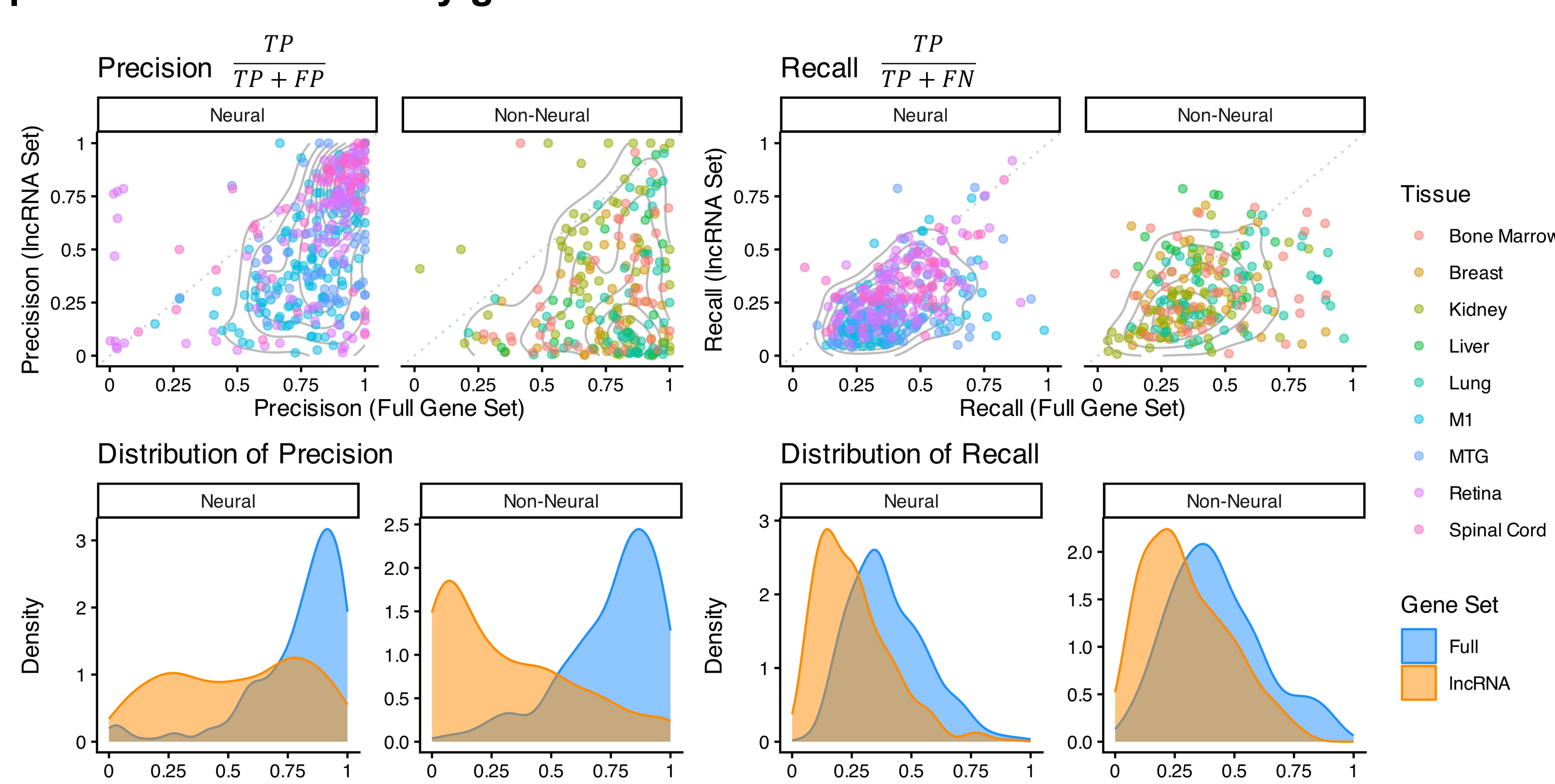


Figure 2. **NS-Forest performance metrics by gene set and neural *versus* non-neural.**

- **Restricting** the gene set to lncRNAs **overall reduces precision** indicating a **loss of information** when removing protein-coding genes.
- **Better precision preservation** in **neural tissues** indicates a greater retention of **discriminatory information** in the lncRNA gene set.



Conclusions

- **Neural tissues** contain **more candidate marker genes** than non-neural tissues, with a **larger contribution from lncRNAs**, suggesting that lncRNAs are more highly expressed in neuronal cells.
- **lncRNAs** exhibit **greater cell type specificity** than the full transcriptome in **neural tissues**.
- Restricting analyses to lncRNAs **generally decreases** the **precision** of NS-Forest classification, but the **reduction is smaller in neural tissues** than in non-neural tissues.
- lncRNAs may play a unique role in **establishing** the **highly specific gene regulation** necessary to generate the large number of **distinct neuron cell subtypes**.
- **Future analyses** of coding-only and downsampled gene sets will help determine and quantify **differences in biomarker potential** between **lncRNA** and **coding genes**.

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